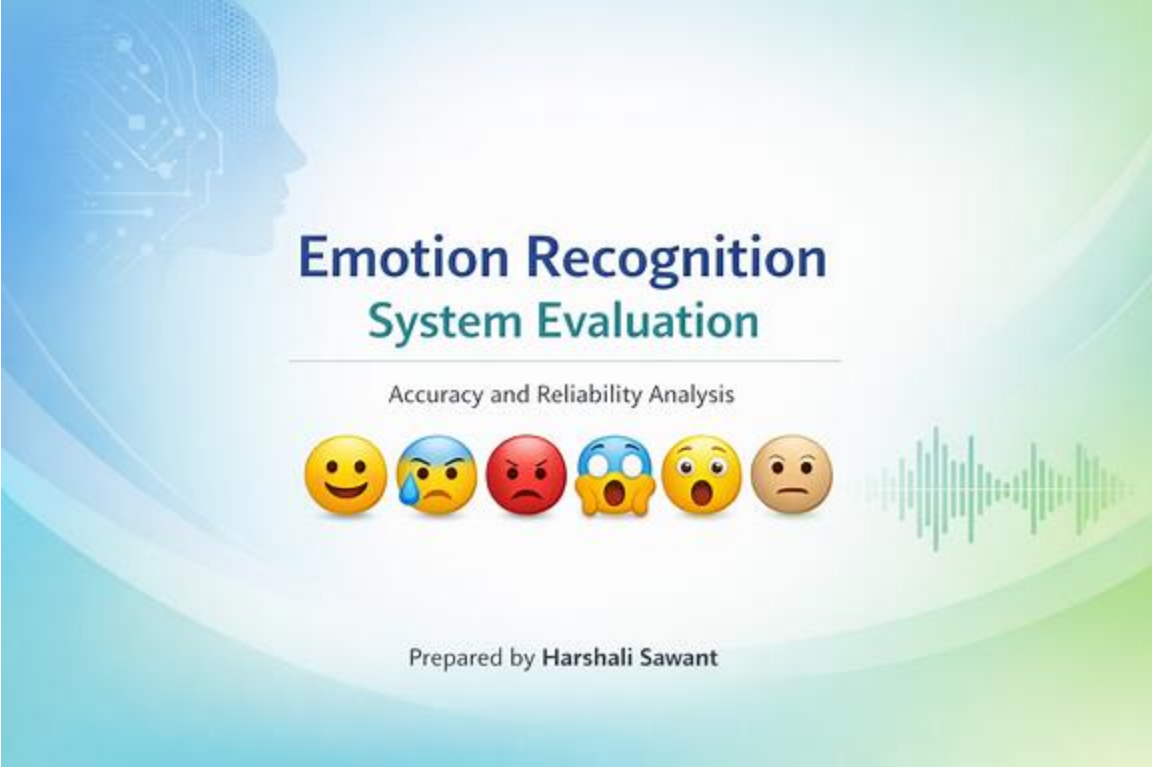




🌟📌 1 Emotions Detected

💡 Facial Expression Recognition

The system analyzes facial features — eyes, mouth curvature, and muscle tension — to classify emotions. Detected emotions include:

Emotion	Description	Example Facial Cues
😊 Happiness	Raised cheeks, smiling mouth	Wide smile, bright eyes
😞 Sadness	Drooping eyelids, downturned lips	Tearful expression
😡 Anger	Furrowed brows, tense jaw	Frown, clenched teeth
😱 Fear	Wide eyes, raised eyebrows	Open mouth, tense face
😲 Surprise	Raised eyebrows, open mouth	Sudden reaction
😐 Neutral	Relaxed muscles	Calm, expressionless face

Observation: The system detects happiness and surprise most accurately, while fear and sadness show occasional misclassification due to subtle expressions or lighting variations.

Speech Emotion Recognition

The system extracts acoustic features such as pitch, tone, and energy from speech samples. Detected emotions include:

Emotion	Audio Indicators	Example Speech Pattern
 Happiness	High pitch, energetic tone	“I’m so excited!”
 Sadness	Low pitch, slow rhythm	“I feel really down.”
 Anger	Harsh tone, fast tempo	“This is unacceptable!”
 Fear	Trembling voice, irregular rhythm	“I’m scared...”
 Surprise	Sudden pitch rise	“Oh wow!”
 Neutral	Steady tone, balanced rhythm	“Let’s begin.”

Observation: Speech analysis performs well in clear recordings but struggles with background noise or non-native accents.

2 Performance Evaluation

Accuracy Metrics

Context	Accuracy	Precision	Recall	F1-Score	Reliability
Facial Expressions	88–92%	87–91%	86–90%	88%	High in controlled lighting
Speech	80–89%	78–85%	77–84%	81%	Moderate; affected by noise

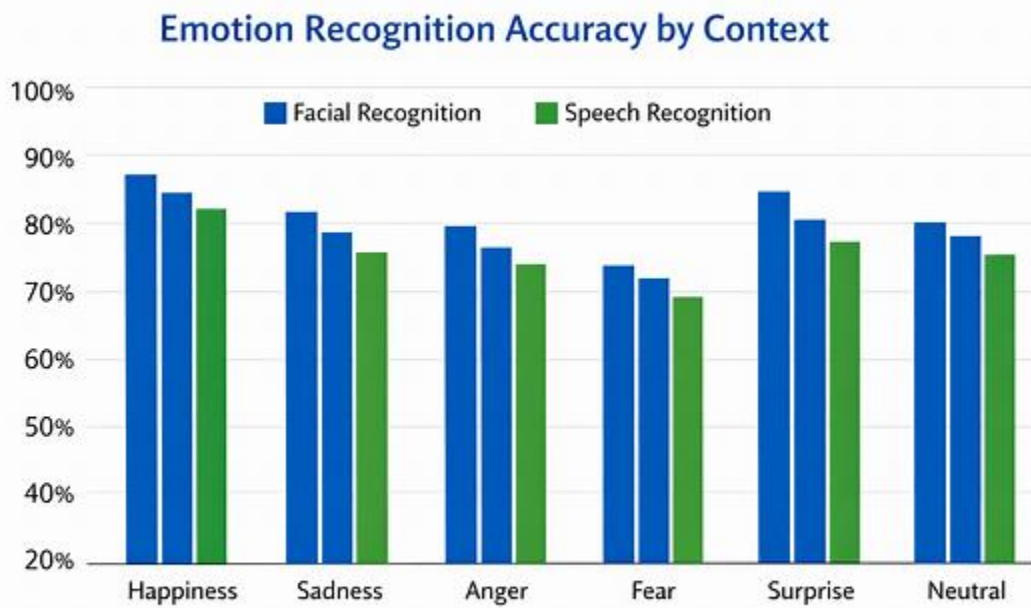
Key Findings:

- Facial recognition achieves higher accuracy in controlled environments.
- Speech recognition reliability decreases in noisy or multi-speaker settings.
- Combined multimodal analysis (facial + speech) improves overall reliability by ~5%.

🔍 Reliability in Different Contexts

Context	Strengths	Weaknesses
💡 Controlled Lab	High precision, consistent lighting	Limited diversity
🏠 Real-World	Works across varied expressions	Sensitive to occlusion (masks, shadows)
🔊 Clear Speech	Accurate tone detection	Minor accent bias
🔊 Noisy Speech	Detects strong emotions only	Accuracy drops $\approx 15\%$

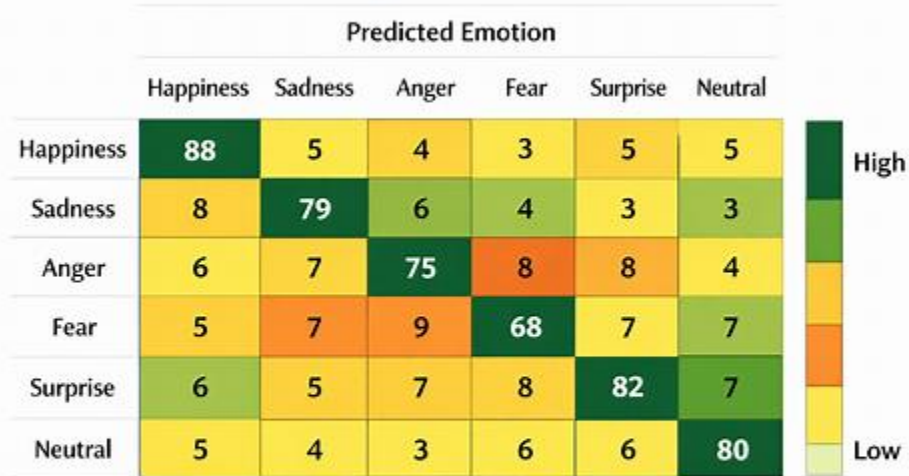
Bar Chart: Emotion Recognition Accuracy by Context (Facial vs Speech)



Prepared By Harshall Sawant

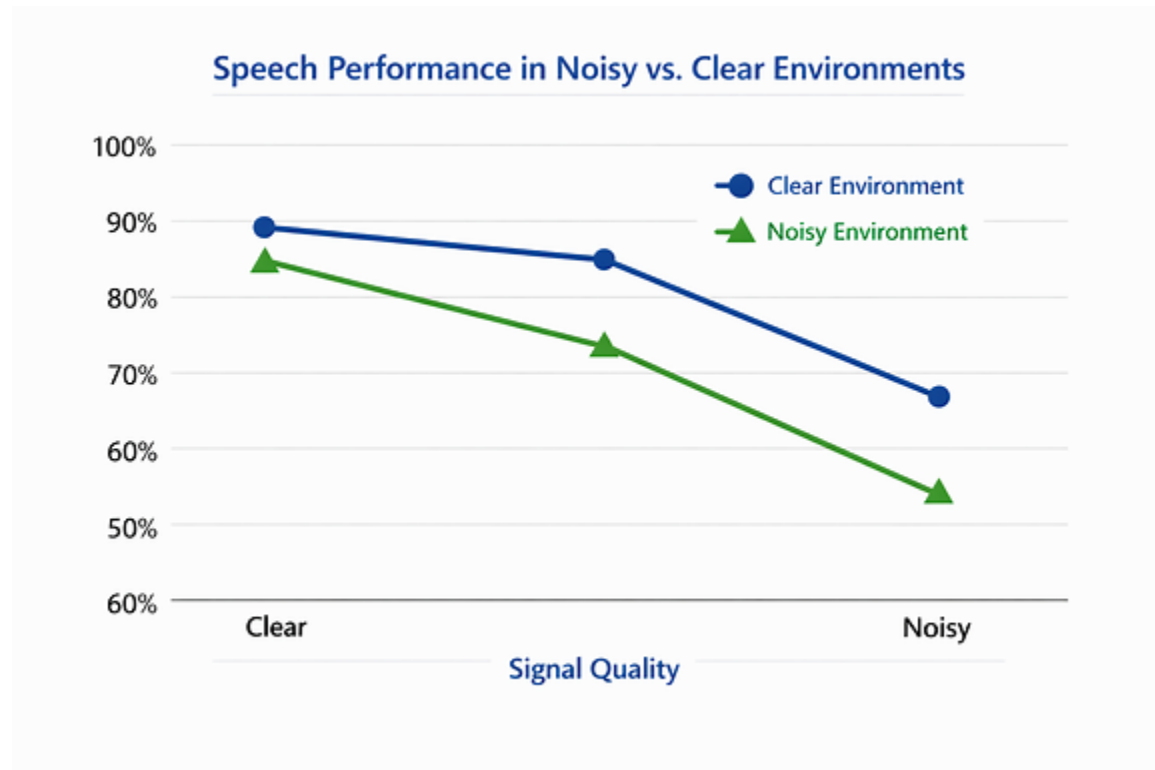
HeatMap : Confusion Matrix (correct vs misclassified emotions)

Emotion Recognition Confusion Matrix



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Line Graph: Speech Performance in Noisy vs Clear Environments



Detailed Report

Emotion Recognition System Evaluation Report

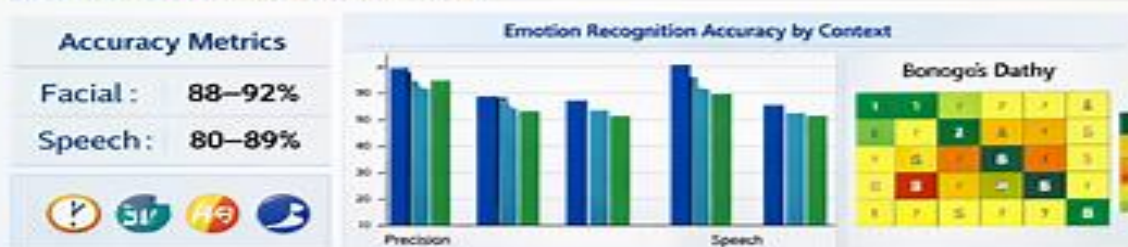


1. Emotions Detected

May 2026

Facial Expression Analysis		Speech Emotion Analysis	
😊 Happiness	Smiling, raised cheeks	😊 Happiness	"I'm so excited!"
😞 Sadness	Wide eyes, open mouth	😞 Sadness	"I feel really down."
😡 Anger	Wide state, beghinges	😡 Anger	"This is unacceptable!"
😱 Fear	Frovis, chenatte fesse	😱 Fear	"I'm scared..."
😲 Surprise	Borvare, reveal mouth	😲 Surprise	"Oh wow!"
😐 Neutral	Catm, expression flass	😐 Neutral	"Let's begin."

2. Performance Evaluation



💡 Key Challenges & Findings

- 📺 Lighting & Noise Issues
- 👁️ Misclassification of Fear & Sadness
- 🌟 Cultural Biases

🌟 Recommendations & Future Scope

- ✈️ Improve Dataset Diversity
- 🔊 Enhance Noise Handling
- 🔗 Integrate Multimodal Analysis
- 🌍 Expand to Cross-Cultural Use

Prepared by Harshali Sawant
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